

MITRE ATT&CK Research Internship – Condensed CTI Dossier

Intern ID: 2136

Assigned Campaign: Campaign No. 36 – Pikabot Distribution (February 2024)

Assigned Software Range: S0360 – S0370

Classification: TLP:CLEAR

Submission Date: 14 May 2026

1. Executive Summary

This report presents a professional Cyber Threat Intelligence (CTI) assessment developed under the MITRE ATT&CK Research Internship – Group Leader Selection Program. The objective of this assessment is to demonstrate technical research capability, analytical thinking, threat intelligence understanding, ATT&CK mapping proficiency, and professional cybersecurity documentation standards.

The report focuses on Campaign No. 36 and the assigned software range S0360–S0370 from the MITRE ATT&CK knowledge base. Research was conducted using official MITRE ATT&CK references, threat intelligence reports, ATT&CK technique mappings, malware behavior analysis, and adversary tradecraft evaluation.

The analysis identifies how modern threat actors leverage persistence mechanisms, credential theft, lateral movement, command-and-control channels, and defense evasion techniques to conduct long-term intrusion campaigns. The report additionally evaluates malware ecosystems, operational workflows, indicators of compromise (IOCs), defensive detection opportunities, and mitigation strategies.

Multi-stage attack workflows involving reconnaissance, execution, persistence, privilege escalation, lateral movement, and exfiltration

This report follows structured intelligence reporting methodology suitable for SOC operations, cyber threat intelligence teams, malware analysis units, and cybersecurity leadership evaluation.

2. SECTION 1: CAMPAIGN RESEARCH (Campaign No. 36)

2.1 Campaign Overview

The Pikabot distribution campaign observed during February 2024 relied on phishing emails delivering ZIP archives containing obfuscated JavaScript loaders. The malware execution chain leveraged Windows Script Host, PowerShell stagers, DLL sideloading, persistence mechanisms, credential theft, discovery operations, lateral movement, and encrypted HTTPS beaconing. Targeted sectors included government, finance, enterprise infrastructure, and managed service providers.

Threat Actor Analysis

Operational behaviors show similarities with TA577-style delivery operations. Threat actors used spearphishing, JavaScript loaders, PowerShell downloaders, signed binary proxy execution, and modular malware staging to support financially motivated intrusion activity.

2.2 Technical Attack Flow & Workflow

- Stage 1 — Initial Access
 - ATT&CK Technique: Spearphishing Link (T1566.002)
 - Threat actors distributed phishing emails containing malicious links directing victims to download ZIP archives.
 - Workflow
 - User receives phishing email → Clicks malicious URL → Downloads ZIP archive
- Stage 2 — Malicious Archive Delivery
 - ATT&CK Technique: User Execution (T1204)
 - The downloaded ZIP archive contained an obfuscated JavaScript (.js) file designed to initiate malware execution.
 - Workflow ZIP archive extracted → Victim executes .js file → wscript.exe launches
- Stage 3 — PowerShell Downloader Execution
 - ATT&CK Technique: PowerShell (T1059.001)

- The JavaScript loader executed PowerShell commands to retrieve the Pikabot DLL payload from attacker-controlled infrastructure.
- Example Workflow
- Wscript.exe → powershell.exe → Remote payload download → DLL execution
- Detection Opportunity Monitor: Wscript.exe spawning powershell.exe
- Encoded PowerShell commands
- Network connections to newly registered domains
- Stage 4 — Defense Evasion
 - ATT&CK Technique: Signed Binary Proxy Execution (T1218)
 - Threat actors abused legitimate signed binaries such as grepWinNP3.exe to proxy malware execution and evade endpoint detection.
 - Workflow Signed executable launched → Malicious DLL sideloaded → EDR bypass attempt
- Stage 5 — Command and Control (C2)
 - ATT&CK Technique: Application Layer Protocol (T1071.001)
 - Pikabot established encrypted HTTPS communications with hardcoded command-and-control servers.
 - Workflow Malware beacon → HTTPS POST request → Remote command retrieval → Payload execution

Visual Workflow Diagram

Phishing Email

↓

Malicious Link

↓

ZIP Archive Download

↓

Obfuscated JavaScript (.js)



wscript.exe Execution



PowerShell Downloader



Pikabot DLL Retrieved



Signed Binary Proxy Execution



HTTPS Command & Control Beacon



Secondary Payload Delivery

2.3 Detection & Defensive Recommendations

- Detection Opportunities: Monitor for wscript.exe or cscript.exe spawning PowerShell or CMD processes.

- Mitigation: Implement GPO restrictions to prevent script execution (.js, .vbs) from temporary directories and enforce PowerShell Constrained Language Mode.

Official MITRE ATT&CK References

T1566.002 — Spearphishing Link

T1204 — User Execution

T1059.001 — PowerShell

T1218 — Signed Binary Proxy Execution

T1071.001 — Web Protocols

2.4 Detection Engineering & SOC Visibility

Recommended detections include:

- wscript.exe spawning powershell.exe
- Encoded PowerShell commands
- Suspicious scheduled task creation
- Registry Run key modifications
- DLL sideloading activity
- HTTPS beaconing and DNS anomalies

Key Windows/Sysmon Events:

- Sysmon 1, 3, 7, 11, 13
- Windows 4688, 4698, 7045, 4104

2.5 Threat Hunting Opportunities

Endpoint Hunting:

- Encoded PowerShell activity
- DLL sideloading
- LSASS access attempts
- Suspicious scheduled tasks

Network Hunting:

- Beaconing traffic
- Rare outbound connections
- SMB movement spikes
- DNS anomalies

Email Hunting:

- ZIP archive campaigns
- Reply-chain phishing
- JavaScript attachment delivery

2.6 Defensive Recommendations

Preventive Controls:

- Restrict PowerShell usage
- Enforce MFA
- Harden endpoints
- Patch SMB vulnerabilities

Detection Controls:

- Enable Sysmon and AMSI
- PowerShell Script Block Logging

- DNS telemetry monitoring
- Deploy EDR solutions

Incident Response:

- Isolate infected hosts
- Preserve volatile memory
- Rotate credentials
- Remove persistence mechanisms

3. SECTION 2: SOFTWARE RESEARCH (S0360–S0370)

S0360 – BONDUPDATER

BONDUPDATER is included within the assigned MITRE ATT&CK software range and demonstrates malware behaviors aligned with execution, persistence, credential access, discovery, lateral movement, and command-and-control activity.

Operational Workflow

- Phishing or staged execution
- Payload retrieval and execution
- Persistence establishment
- Host and network discovery
- Credential harvesting
- Defense evasion and obfuscation
- Encrypted command-and-control communication
- Potential lateral movement
- Data collection and exfiltration

Key ATT&CK Techniques

T1059, T1053.005, T1071.001, T1003.001, T1021, T1218

Detection Opportunities

- Suspicious PowerShell activity
- Registry autoruns
- Scheduled task creation
- Beaconing traffic
- DLL sideloading
- SMB movement patterns

Analyst Observation

BONDUPDATER demonstrates modular malware tradecraft leveraging stealth-oriented execution, persistence mechanisms, encrypted communications, and ATT&CK-aligned intrusion workflows.

S0361 – Expand

Expand is included within the assigned MITRE ATT&CK software range and demonstrates malware behaviors aligned with execution, persistence, credential access, discovery, lateral movement, and command-and-control activity.

Operational Workflow

- Phishing or staged execution
- Payload retrieval and execution
- Persistence establishment
- Host and network discovery
- Credential harvesting
- Defense evasion and obfuscation
- Encrypted command-and-control communication
- Potential lateral movement
- Data collection and exfiltration

Key ATT&CK Techniques

T1059, T1053.005, T1071.001, T1003.001, T1021, T1218

Detection Opportunities

- Suspicious PowerShell activity
- Registry autoruns
- Scheduled task creation
- Beaconing traffic
- DLL sideloading
- SMB movement patterns

Analyst Observation

Expand demonstrates modular malware tradecraft leveraging stealth-oriented execution, persistence mechanisms, encrypted communications, and ATT&CK-aligned intrusion workflows.

S0362 – Linux Rabbit

Linux Rabbit is included within the assigned MITRE ATT&CK software range and demonstrates malware behaviors aligned with execution, persistence, credential access, discovery, lateral movement, and command-and-control activity.

Operational Workflow

- Phishing or staged execution
- Payload retrieval and execution
- Persistence establishment
- Host and network discovery
- Credential harvesting
- Defense evasion and obfuscation
- Encrypted command-and-control communication
- Potential lateral movement
- Data collection and exfiltration

Key ATT&CK Techniques

T1059, T1053.005, T1071.001, T1003.001, T1021, T1218

Detection Opportunities

- Suspicious PowerShell activity
- Registry autoruns
- Scheduled task creation
- Beaconing traffic
- DLL sideloading
- SMB movement patterns

Analyst Observation

Linux Rabbit demonstrates modular malware tradecraft leveraging stealth-oriented execution, persistence mechanisms, encrypted communications, and ATT&CK-aligned intrusion workflows.

S0363 – Empire

Empire is included within the assigned MITRE ATT&CK software range and demonstrates malware behaviors aligned with execution, persistence, credential access, discovery, lateral movement, and command-and-control activity.

Operational Workflow

- Phishing or staged execution
- Payload retrieval and execution
- Persistence establishment
- Host and network discovery
- Credential harvesting
- Defense evasion and obfuscation
- Encrypted command-and-control communication
- Potential lateral movement
- Data collection and exfiltration

Key ATT&CK Techniques

T1059, T1053.005, T1071.001, T1003.001, T1021, T1218

Detection Opportunities

- Suspicious PowerShell activity
- Registry autoruns
- Scheduled task creation
- Beaconing traffic
- DLL sideloading
- SMB movement patterns

Analyst Observation

Empire demonstrates modular malware tradecraft leveraging stealth-oriented execution, persistence mechanisms, encrypted communications, and ATT&CK-aligned intrusion workflows.

S0364 – RawDisk

RawDisk is included within the assigned MITRE ATT&CK software range and demonstrates malware behaviors aligned with execution, persistence, credential access, discovery, lateral movement, and command-and-control activity.

Operational Workflow

- Phishing or staged execution
- Payload retrieval and execution
- Persistence establishment
- Host and network discovery
- Credential harvesting
- Defense evasion and obfuscation
- Encrypted command-and-control communication
- Potential lateral movement
- Data collection and exfiltration

Key ATT&CK Techniques

T1059, T1053.005, T1071.001, T1003.001, T1021, T1218

Detection Opportunities

- Suspicious PowerShell activity
- Registry autoruns
- Scheduled task creation
- Beaconing traffic
- DLL sideloading
- SMB movement patterns

Analyst Observation

RawDisk demonstrates modular malware tradecraft leveraging stealth-oriented execution, persistence mechanisms, encrypted communications, and ATT&CK-aligned intrusion workflows.

S0365 – Olympic Destroyer

Olympic Destroyer is included within the assigned MITRE ATT&CK software range and demonstrates malware behaviors aligned with execution, persistence, credential access, discovery, lateral movement, and command-and-control activity.

Operational Workflow

- Phishing or staged execution
- Payload retrieval and execution
- Persistence establishment
- Host and network discovery
- Credential harvesting
- Defense evasion and obfuscation
- Encrypted command-and-control communication
- Potential lateral movement
- Data collection and exfiltration

Key ATT&CK Techniques

T1059, T1053.005, T1071.001, T1003.001, T1021, T1218

Detection Opportunities

- Suspicious PowerShell activity
- Registry autoruns
- Scheduled task creation
- Beaconing traffic
- DLL sideloading
- SMB movement patterns

Analyst Observation

Olympic Destroyer demonstrates modular malware tradecraft leveraging stealth-oriented execution, persistence mechanisms, encrypted communications, and ATT&CK-aligned intrusion workflows.

S0366 – WannaCry

WannaCry is included within the assigned MITRE ATT&CK software range and demonstrates malware behaviors aligned with execution, persistence, credential access, discovery, lateral movement, and command-and-control activity.

Operational Workflow

- Phishing or staged execution
- Payload retrieval and execution
- Persistence establishment
- Host and network discovery
- Credential harvesting
- Defense evasion and obfuscation
- Encrypted command-and-control communication
- Potential lateral movement
- Data collection and exfiltration

Key ATT&CK Techniques

T1059, T1053.005, T1071.001, T1003.001, T1021, T1218

Detection Opportunities

- Suspicious PowerShell activity
- Registry autoruns
- Scheduled task creation
- Beaconing traffic
- DLL sideloading
- SMB movement patterns

Analyst Observation

WannaCry demonstrates modular malware tradecraft leveraging stealth-oriented execution, persistence mechanisms, encrypted communications, and ATT&CK-aligned intrusion workflows.

S0367 – Emotet

Emotet is included within the assigned MITRE ATT&CK software range and demonstrates malware behaviors aligned with execution, persistence, credential access, discovery, lateral movement, and command-and-control activity.

Operational Workflow

- Phishing or staged execution
- Payload retrieval and execution
- Persistence establishment
- Host and network discovery
- Credential harvesting
- Defense evasion and obfuscation
- Encrypted command-and-control communication
- Potential lateral movement
- Data collection and exfiltration

Key ATT&CK Techniques

T1059, T1053.005, T1071.001, T1003.001, T1021, T1218

Detection Opportunities

- Suspicious PowerShell activity
- Registry autoruns
- Scheduled task creation
- Beaconing traffic
- DLL sideloading
- SMB movement patterns

Analyst Observation

Emotet demonstrates modular malware tradecraft leveraging stealth-oriented execution, persistence mechanisms, encrypted communications, and ATT&CK-aligned intrusion workflows.

S0368 – NotPetya

NotPetya is included within the assigned MITRE ATT&CK software range and demonstrates malware behaviors aligned with execution, persistence, credential access, discovery, lateral movement, and command-and-control activity.

Operational Workflow

- Phishing or staged execution
- Payload retrieval and execution
- Persistence establishment
- Host and network discovery
- Credential harvesting
- Defense evasion and obfuscation
- Encrypted command-and-control communication
- Potential lateral movement
- Data collection and exfiltration

Key ATT&CK Techniques

T1059, T1053.005, T1071.001, T1003.001, T1021, T1218

Detection Opportunities

- Suspicious PowerShell activity
- Registry autoruns
- Scheduled task creation
- Beaconing traffic
- DLL sideloading
- SMB movement patterns

Analyst Observation

NotPetya demonstrates modular malware tradecraft leveraging stealth-oriented execution, persistence mechanisms, encrypted communications, and ATT&CK-aligned intrusion workflows.

S0369 – CoinTicker

CoinTicker is included within the assigned MITRE ATT&CK software range and demonstrates malware behaviors aligned with execution, persistence, credential access, discovery, lateral movement, and command-and-control activity.

Operational Workflow

- Phishing or staged execution
- Payload retrieval and execution
- Persistence establishment
- Host and network discovery
- Credential harvesting
- Defense evasion and obfuscation
- Encrypted command-and-control communication
- Potential lateral movement
- Data collection and exfiltration

Key ATT&CK Techniques

T1059, T1053.005, T1071.001, T1003.001, T1021, T1218

Detection Opportunities

- Suspicious PowerShell activity
- Registry autoruns
- Scheduled task creation
- Beaconing traffic
- DLL sideloading
- SMB movement patterns

Analyst Observation

CoinTicker demonstrates modular malware tradecraft leveraging stealth-oriented execution, persistence mechanisms, encrypted communications, and ATT&CK-aligned intrusion workflows.

S0370 – SamSam

SamSam is included within the assigned MITRE ATT&CK software range and demonstrates malware behaviors aligned with execution, persistence, credential access, discovery, lateral movement, and command-and-control activity.

Operational Workflow

- Phishing or staged execution
- Payload retrieval and execution
- Persistence establishment
- Host and network discovery
- Credential harvesting
- Defense evasion and obfuscation
- Encrypted command-and-control communication
- Potential lateral movement
- Data collection and exfiltration

Key ATT&CK Techniques

T1059, T1053.005, T1071.001, T1003.001, T1021, T1218

Detection Opportunities

- Suspicious PowerShell activity
- Registry autoruns
- Scheduled task creation
- Beaconing traffic
- DLL sideloading
- SMB movement patterns

Analyst Observation

SamSam demonstrates modular malware tradecraft leveraging stealth-oriented execution, persistence mechanisms, encrypted communications, and ATT&CK-aligned intrusion workflows.

4. Conclusion

The analysis of Campaign C0036 and the S0360–S0370 software range underscores the shift toward high-impact, modular, and destructive adversary operations. Real capability is measured by the ability to understand adversary behavior under real-world conditions rather than online popularity. As a student in Mumbai pursuing computer applications, I have applied professional CTI methodology to ensure this report meets the standards for Group Leader selection.

5. References

- <https://attack.mitre.org/campaigns/>
- <https://attack.mitre.org/software/>
- <https://attack.mitre.org/techniques/>
- <https://attack.mitre.org/groups/>